6-DOF Physical AI Robotic Manipulator

Autonomous Motion Planning & Real-Time Embedded Control

PRODUCTION GRADE

Domain: Physical AI / Robotics
Platform: FreeRTOS / ESP32

< 15 ms	50 Hz	6-DOF	100%
IK CONVERGENCE	PWM DETERMINISM	KINEMATIC CHAIN	ON-DEVICE EDGE AI

1. Executive Summary & 30-Second Elevator Pitch

> *"I engineered a 6-DOF physical-AI robotic manipulator for autonomous spatial manipulation and trajectory execution. Running on an embedded 32-bit MCU under real-time FreeRTOS, the architecture coordinates six high-torque metal-gear digital servos through hardware timers at 50 Hz. Featuring an analytical inverse kinematics solver converging in under 15ms and cubic-spline S-curve interpolation, the system achieves sub-millimeter repeatable convergence with zero cloud latency."*

Firmware & Motion Architecture

- **FreeRTOS Task Scheduler**: Dedicated 50 Hz motion planning task with microsecond hardware timer determinism.
- **Analytical IK Solver**: Closed-form Denavit-Hartenberg (DH) matrix transformation in <15 ms.
- **S-Curve Trajectory Limiting**: Cubic Hermite splines to suppress joint inertia and prevent mechanical gear lash.

Hardware & Mechatronics

- **Actuation**: 6x High-Torque Metal-Gear digital servos mounted on CNC reinforced aluminum alloy brackets.
- **Power Topology**: Dual isolated rails (5V/10A servo bus with transient TVS protection).
- **End-Effector**: Dual-gear compliance claw with proportional gripping telemetry.

2. Actuation & Joint Telemetry Specifications

Joint Axis	Actuator Type	Operating Range	Control Interface	Torque Rating
Joint 1 (Base Yaw)	Metal-Gear Digital Servo	0° to 180°	Hardware PWM (50 Hz)	20 kg·cm @ 6.0V
Joint 2 (Shoulder Pitch)	Metal-Gear Digital Servo	15° to 165°	Hardware PWM (50 Hz)	25 kg·cm @ 6.0V
Joint 3 (Elbow Pitch)	Metal-Gear Digital Servo	0° to 170°	Hardware PWM (50 Hz)	20 kg·cm @ 6.0V
Joint 4 (Wrist Pitch)	Digital Micro-Servo	0° to 180°	Hardware PWM (50 Hz)	15 kg·cm @ 6.0V
Joint 5 (Wrist Roll)	Digital Micro-Servo	0° to 180°	Hardware PWM (50 Hz)	15 kg·cm @ 6.0V
Joint 6 (End-Effector Claw)	Dual-Gear Linear Gripper	0 to 55 mm Stroke	Hardware PWM (50 Hz)	Proportional Compliance